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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21313

F-413

B.C.A. (IIIrd Semester) Examination, 2019-20

(New Course)

DATA STRUCTURES USING C

Paper-BCA-303-N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) What is data structure ? Discuss the different types of data structures with features.

   (b) What is Stack ? Write an algorithm to implement stack using linked list.

2. (a) Explain why circular queue is better than linear queue.

   (b) Write a function to delete a node from beginning of double linked list.

3. (a) Write a recursive function to count number of nodes in tree.

   (b) Convert the following infix expression into post-fix expression.

   (A + B – C * D)/H

4. (a) Compare Selection sort and Merge sort.

   (b) Define Hashing. Discuss any *two* hashing methods with example.

5. (a) Explain height balance tree and AVL. tree with suitable example.

   (b) Explain the concept of asymptotic notations. Explain the different asymptotic natations.

6. (a) Construct a binary tree for the following inorder traversal :

   B G H D A E C I J K F J

   (b) Explain Overflow and Underflow condition on stack.

7. (a) What is BST ? Explain algorithm for insertion in BST.

   (b) What is Graphs ? Explain Multigraphs and Directed Graphs

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8. Explain any *four* of the following :

(i) D-queue and Priority queue

(ii) Doubly Linked list

(iii) Tower of Hanoi problem with *three* disc

(iv) Sequential and Binary Search

(v) Sparse Matrix.

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